

Target-Weighted Experimental Sample Design for Predicting Unit-Specific Treatment Effects

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ABSTRACT. Randomized experiments are increasingly used to predict treatment effects for units the study never enrolled, and whom to enroll then matters as much as how many. Tipton and Mamakos showed that the prediction risk of an experiment depends on the sample through a single computable mismatch factor, and that a deliberately heterogeneous sample can predict better than a representative one. This article turns that finding into a design problem: given the target, how should the sample be chosen? Under the trusted working model the answers are constructive. Matching the target’s covariate means is always advantageous; a budget of covariate spread is best allocated in proportion to the matrix square root of the target covariance rather than the covariance itself; and the choice of enrollment shares across candidate sites is a convex program whose certified optimum uses few sites, while enrolling the sites most similar to the target proves worst in our worked example. The same calculus, however, rewards unlimited spread, so we ask what survives when the linear model is only approximately trusted. Bounding the curvature of the treatment-effect contrast restores an interior optimum and removes the square-root tilt: the criterion-optimal covariance has the target’s own shape, inflated by a factor that fades as samples grow, as the curvature bound rises, and as moderators accumulate. Representativeness thus reappears as the cautious limit of design rather than its starting point. The criterion is not a complete account of total risk, and we price its omissions exactly; noise-adjusted and exact finite-sample comparisons complete the toolkit.

Keywords: experimental design; generalizability; covariate shift; site selection; optimal design; robustness; unit-specific treatment effects

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1. Introduction

Randomized experiments are increasingly asked to support predictions for populations they did not sample. A state-level evaluation informs a national rollout, a pilot in volunteer districts informs a policy for all districts, and in each case the units whose treatment effects must be predicted are not the units that were enrolled. Tipton and Mamakos (2025) showed that, when the goal is unit-specific treatment-effect (USTE) prediction, this mismatch is not a vague external-validity concern but a computable quantity: the disagreement between the experimental sample and the prediction target enters the risk through a single factor that depends only on the first two covariate moments of each, and the factor separates into a mean-shift component and a dispersion-comparison component. Moreover, as they emphasize, the factor is not a one-directional penalty. Deliberately maximizing covariate heterogeneity in the sample can push prediction error below that of a random sample from the target; in their school-level application, an eighty-school state evaluation would have needed roughly 113 schools, after reweighting, to match the precision of eighty schools drawn nationally. Companion work (Lee, 2026) expresses their factor as a relative leverage, gives its exact finite-sample form for Gaussian designs, and shows that the same object governs when tuned analysis of the resulting data is guaranteed to be safe.

Those results price a sample that has already been chosen. In this article we take up the design question they leave open: given a target population, how should the experimental sample be chosen in the first place? The question arises in three practical forms, and we treat all three. A designer who can shape the recruitment distribution must decide where to place covariate spread. A designer who must assemble the sample from a pool of candidate sites must decide which sites to enroll and in what proportions. And every designer must decide how far to trust the working model when the sample deliberately departs from the target, because the case for departing rests entirely on that model.

Under the trusted working model, the answers turn out to be simple enough to state in words. At any fixed covariate dispersion, matching the target’s covariate means is always advantageous, and everything difficult about the design lives in the covariance. When a total budget of covariate spread must be allocated across moderator directions, the best allocation follows the matrix square root of the target covariance rather than the covariance itself, so the optimal sample is deliberately flatter than a representative one, and the representative shape is optimal only when the target is isotropic. When the sample must instead be assembled from candidate sites, the

choice of enrollment shares is a convex program: it can be solved reliably, the solution can be certified optimal, and the optimum requires only a small number of sites, bounded by the dimension of the moment matrix rather than by the size of the pool. In the thirty-site example we carry through the article, the certified optimum enrolls four sites, whereas the intuitive alternative of enrolling the five sites that look most similar to the target is the worst design we evaluate, costing more than twice the achievable leverage. The reason is that a cluster of near-target sites is typically an underdispersed source, and proximity in means does nothing to supply the spread the prediction problem needs.

These prescriptions share a worrying feature: taken literally, the trusted-model calculus rewards unbounded covariate spread, since an ever-wider sample makes regression estimates ever more precise wherever the linear model is exact. No practitioner would follow that advice, and the reason is model doubt rather than cost alone. The second half of the article therefore asks what design guidance survives when the working model is only approximately right. We bound the curvature of the treatment-effect contrast and minimize an explicit criterion that combines the first-order estimation risk with the worst-case extrapolation bias the bound permits. The bound restores an interior optimum, and in several dimensions it overturns the square-root allocation: the criterion-optimal covariance has the target’s own shape, inflated by a single scalar that decreases toward one as the sample grows, as the curvature bound rises, and as moderators accumulate. In this specific sense representativeness reemerges as a limiting case rather than a starting assumption. We are careful, however, about what the criterion does and does not control. It is not a complete account of total risk, and we price its omissions exactly; under strong curvature the complete first-order optimum can even fall below the representative spread, so deliberate overdispersion is a property of the stated criterion and not a universal prescription. Figure 1 summarizes the resulting structure of the article: a trusted-model column whose calculus is constructive but unbounded, a bounded-curvature column that supplies the missing counterweight, and a comparison layer through which any final choice among candidate designs should pass.

It is worth stating clearly what this article does not claim. The leverage criterion is classical linear optimality with the target second moment as the weight matrix, in the lineage of Kiefer and Wolfowitz (1960), Kiefer (1974), Fedorov (1972), and Pukelsheim (1993), and the tension between variance and extrapolation bias under model inadequacy has been response-surface-design territory since Box and Draper

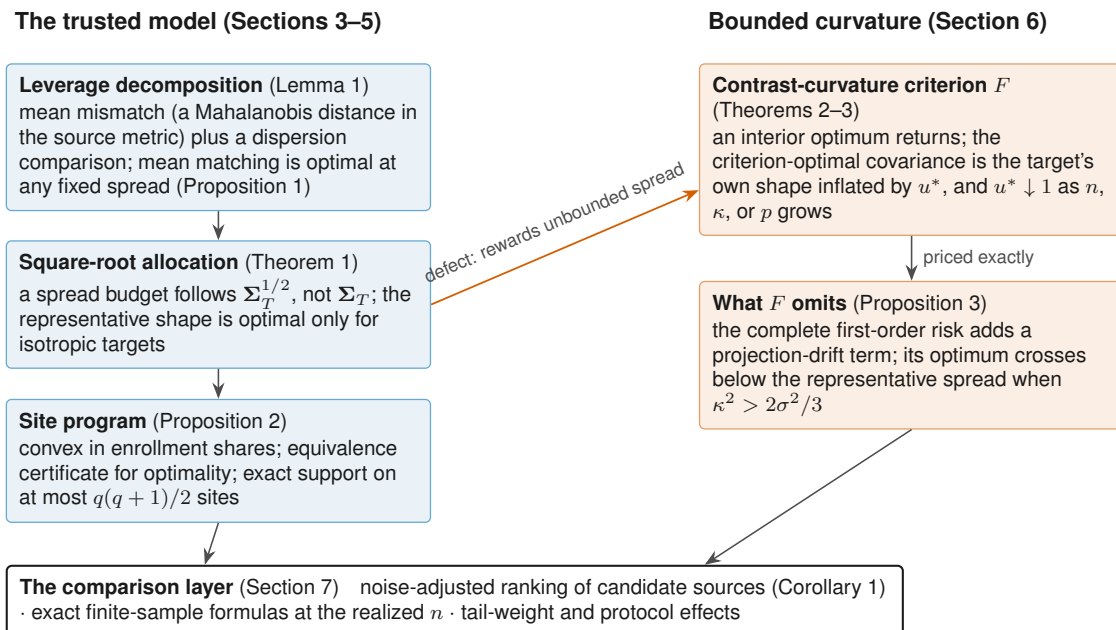


FIGURE 1. Overview of the article. The left column develops the trusted-model calculus: the leverage decomposition and mean matching (Section 3), the square-root allocation under a spread budget (Section 4), and the site-selection program with its certificates (Section 5). Its defect, the reward to unbounded spread, motivates the right column: a bounded contrast-curvature criterion whose optimum has the target’s own shape, moderately inflated (Section 6), together with an exact account of what the criterion omits. Both columns feed the comparison layer of Section 7, where candidate designs are ranked with noise adjustments and exact finite-sample formulas.

(1959). Our contribution is the specialization to target-population USTE prediction: the target supplies the metric, a spectral bound on contrast curvature yields a nuclear-norm penalty, and the resulting problem has a closed-form solution whose dimension comparative static, as far as we are aware, has not been pointed out before. The decomposition of the mismatch factor and the possibility of beating the representative design are results of Tipton and Mamakos (2025); what we add is the conversion of those facts into allocation rules, an implementable site program, and a scoped robustness criterion. All results are statements within the design model: randomization protects the internal comparison, but predictions for a non-enrolled target rest on the working model across the combined covariate range. We do not treat observational transport, confounded sampling, or outcome-dependent site participation (Degtiar & Rose, 2023; Stuart et al., 2011); sites here differ in covariates, noise, and cost, not

in unmeasured ways. Analysis-side tuning is the subject of the companion article (Lee, 2026) and is complementary to sample-side design. Finally, the residual floor of the prediction risk, with its non-identified potential-outcome correlation, is inherited unchanged, because nothing about the choice of source can relax it.

The plan of the article is as follows. In Section 2 we set out the prediction-risk framework and the relative-leverage factor and position the design question among the generalizability, optimal-design, and site-selection literatures. Section 3 formalizes the design problem and establishes the mean-matching result. Section 4 derives the square-root allocation under a spread budget, and Section 5 develops site selection as a convex program with an exact support certificate. Section 6 introduces the contrast-curvature criterion, first for one moderator and then in general, and prices exactly what the criterion omits. Section 7 adjusts the comparison for noise differences between candidate sources and for finite-sample effects that the first-order calculus cannot see. Section 8 concludes with practical recommendations and limitations. Proofs, derivations, and a computational verification suite covering every formal claim appear in Appendices A–E.

2. Prediction Risk for a Target Population

2.1. The framework and the leverage factor. We inherit the conditional design framework of Lee (2026), built on Tipton and Mamakos (2025). Units carry potential outcomes and moderators $\mathbf{x} \in \mathbb{R}^p$, augmented as $\tilde{\mathbf{x}} = (1, \mathbf{x}^\top)^\top$ with $q = p + 1$; a balanced two-arm experiment of n units per arm is drawn from a source population S ; and USTEs are predicted for a target population T through an arm-wise Gaussian linear working model with source residual scales $\sigma_{a,S}$ held fixed. Writing $\mathbf{M}_P = \mathbb{E}_P(\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top)$ for the augmented second moment of population P , the first-order target risk is

$$R_T(S) = \left(\frac{\sigma_{0,S}^2 + \sigma_{1,S}^2}{n} \right) L_{T|S} + \tau_T^2(\rho_T), \quad L_{T|S} = \text{tr}(\mathbf{M}_S^{-1} \mathbf{M}_T), \quad (1)$$

where the floor $\tau_T^2(\rho_T)$ collects residual effect heterogeneity and is untouched by any choice studied in this article. We shall refer to $L_{T|S}$ as the relative leverage of target T over source S ; it is the only channel through which the design enters the estimation component of the risk. The companion’s exact Gaussian-design refinement replaces $L_{T|S}/n$ by $a_n + b_n(L_{T|S} - 1)$ with known constants, and we shall use that refinement for the small-sample comparisons of Section 7.

Three boundary facts, the first two established by Tipton and Mamakos (2025) and restated in leverage form by the companion, frame what follows. Leverage equals q whenever the source matches the target’s first two moments, so q is the representative benchmark. No inequality of the form $L_{T|S} \geq q$ holds, however: leverage falls below q for an overdispersed source, and it rises far above q when the source is narrow in even one moderator direction (a source of this kind reached 11.25 against $q = 3$ in the companion’s scenarios), so design has real room to improve on representativeness and real room to do harm. Finally, leverage multiplies the source residual variances in (1), so a leverage ranking of candidate sources and a risk ranking can disagree when the candidates differ in noise; we shall return to this in Section 7.1.

2.2. Neighboring literatures. The design question sits among three literatures. In the generalizability tradition, sample selection for external validity is a developed practice: Stuart et al. (2011) and Tipton (2014a) assess similarity between sample and population through propensity-type indices, Tipton (2013) and Tipton (2014b) select stratified samples toward population composition, Litwok et al. (2023) study district and school recruitment strategies by simulation, Tipton (2021) designs trials so that treatment-effect heterogeneity itself is estimable, and the transported-prediction literature supplies the associated weighting machinery (Degtiar & Rose, 2023; Shimodaira, 2000; Steingrimsson et al., 2023). This tradition has largely treated representativeness as the aspiration. Tipton and Mamakos (2025) showed that for prediction it is better understood as a benchmark, and this article asks what the optimum actually looks like. In the optimal-design tradition, criteria of the form $\text{tr}(\mathbf{M}^{-1}\mathbf{A})$ are classical linear optimality, with equivalence theorems and support bounds available since Kiefer and Wolfowitz (1960), Kiefer (1974), Fedorov (1972), and Pukelsheim (1993). Model-robust response-surface design is equally long-standing: Box and Draper (1959) minimized integrated prediction error when a fitted first-degree surface may face quadratic truth, Steinberg (1985) chose the scale of two-level factorials under explicit model inadequacy, and Wiens (1992, 1994) derived minimax designs against bounded departures from the linear model. Our narrower contribution is to make the target population the metric for USTE design and to work out the consequences: the square-root allocation, the site program with exact certificates, and the target-whitened contrast-curvature criterion with its dimension effect.

The third literature is current work on experiment-site selection, and because a recent paper addresses a closely related problem, we position against it and its neighbors explicitly. Table 1 compares the methods by estimand, design variable, shift model,

robustness device, and algorithm. Bouyamourn (2025) selects sites under distribution shift by minimizing optimal-transport bounds on population-average and conditional-average treatment-effect error, adding a Wasserstein distributionally robust program when the deployment distribution is itself uncertain. Tang et al. (2026) propose the R-IDEA acquisition function, which trades representativeness, informativeness, and error de-amplification for Bayesian active learning under model misspecification. Shi and Lin (2022) choose experimental samples from an observational pool to sharpen a combined estimator of the population average effect, and Ouyang et al. (2016) jointly select patients and allocate treatment to estimate covariate-dependent effects efficiently. None of these targets unit-specific effect-prediction risk together with its inherited residual floor, and none uses the target-whitened contrast-curvature criterion developed here. Conversely, we do not treat transport-metric ambiguity sets, so a designer whose primary worry is distributional shift beyond second moments should read Section 6 as complementary to that machinery rather than competing with it.

3. The Design Problem and Mean Matching

For first-order purposes the design variable is the source distribution itself, summarized by its moderator mean $\boldsymbol{\mu}_S$ and covariance $\boldsymbol{\Sigma}_S$; the target contributes $\boldsymbol{\mu}_T$ and $\boldsymbol{\Sigma}_T$. The first task is to understand how these four objects determine the leverage factor, and the answer is an exact decomposition.

Lemma 1 (Leverage decomposition). With augmented second moments as above and $\boldsymbol{\Sigma}_S \succ \mathbf{0}$,

$$L_{T|S} = 1 + (\boldsymbol{\mu}_T - \boldsymbol{\mu}_S)^\top \boldsymbol{\Sigma}_S^{-1} (\boldsymbol{\mu}_T - \boldsymbol{\mu}_S) + \text{tr}(\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\Sigma}_T). \quad (2)$$

The proof, an exercise in block inversion, is given in Appendix A; the decomposition is the mean-shift and dispersion structure of Tipton and Mamakos (2025) restated in augmented-moment form. Leverage separates into an intercept unit, a mean-mismatch term that is a Mahalanobis distance in the source metric, and a spread term that compares the two covariances direction by direction. Note that the source covariance appears inversely in both non-constant terms: source spread simultaneously discounts any mean mismatch and shrinks the dispersion comparison. Figure 2 displays the resulting geometry for one moderator, and the whole trusted-model calculus of the next two sections can be read from its contours. Mean mismatch is punished at every spread, more severely the narrower the source; and at any fixed

TABLE 1. Position relative to current sample- and site-selection methods. Rows are ordered by publication date; the last row is this article.

Method	Estimand	Design variable	Shift / robustness device	Algorithm
Ouyang et al. (2016)	covariate-dependent treatment effects	patient selection and arm allocation	model-based DOE	selection/allocation optimization
Shi and Lin (2022)	population ATE via a combined estimator	RCT sample from an observational pool	none (trusted outcome model)	covariate-allocation strategy
Litwok et al. (2023)	average effect, generalizability	site recruitment	population frame	ranked recruitment
Bouyamourn (2025)	PATE and CATE error under shift	site selection	optimal transport and Wasserstein DRO	transport and DRO programs
Tang et al. (2026)	test-distribution prediction	experimental-input acquisition	model misspecification and covariate shift	R-IDEA acquisition
This article	unit-specific effect-prediction risk	source moments and site weights	bounded contrast-curvature criterion	closed forms; convex program with equivalence and exact support certificates

mean the leverage falls monotonically as the source widens, a feature to which we return with some suspicion in Section 6.

The mean statement deserves to be made precise, because it is the one unconditional prescription the calculus offers.

Proposition 1 (Mean matching at fixed covariance). For any fixed $\Sigma_S \succ \mathbf{0}$, the leverage (2) is uniquely minimized over μ_S at $\mu_S = \mu_T$.

Proposition 1 is elementary, and its scope deserves as much attention as its content. At a fixed source covariance the mean decision has an unconditional answer, so everything difficult about source design lives in the covariance; this is why the remainder of the article is almost entirely about spread. However, the proposition is a directional diagnostic rather than a joint-design command. In site-assembly problems

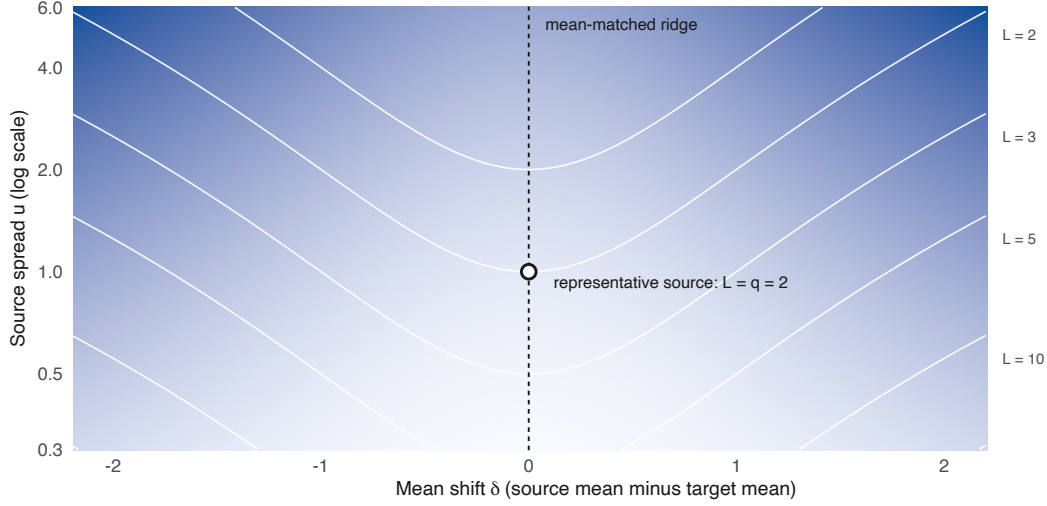


FIGURE 2. The trusted-model design calculus for one moderator, drawn from the exact decomposition (2). Contours show the leverage L of a source with mean shift δ and spread u against a standardized target. Leverage is minimized in δ along the mean-matched ridge $\delta = 0$ (Proposition 1), declines monotonically in u at every δ , and equals the representative benchmark $q = 2$ at the point $(0, 1)$. The dashed path marks the mean-matched slice $L = 1 + 1/u$, which has no interior minimum; Section 6 supplies the missing counterweight.

the enrollment weights move means, covariances, noise, and cost together, the target mean need not even lie in the feasible convex hull, and the correct object to optimize is the full mixture moment of Section 5, which prices mean and spread jointly. When mean matching is infeasible or deliberately incomplete, the residual mismatch $\delta = \mu_T - \mu_S$ simply joins the spread problem: the decomposition can be rewritten as $1 + \text{tr}\{\Sigma_S^{-1}(\Sigma_T + \delta\delta^\top)\}$, and the allocation theory of the next section applies with the augmented weight matrix $\mathbf{A} = \Sigma_T + \delta\delta^\top$.

4. Allocating Spread: The Square-Root Rule

Suppose then that means are matched, or that the residual mismatch has been absorbed into the weight matrix $\mathbf{A}$, and that the designer controls the source covariance subject to a budget on total spread. The budget makes the problem well posed, and its solution has a form that we did not find obvious in advance.

Theorem 1 (Square-root allocation). Let $\mathbf{A} = \Sigma_T + \delta\delta^\top \succ \mathbf{0}$ and consider minimizing $L_{T|S}$ over source covariances with a spread budget, $\Sigma_S \succ \mathbf{0}$ with $\text{tr}(\Sigma_S) \leq$

B . Then: (i) the unique minimizer and minimum are

$$\boldsymbol{\Sigma}_S^* = B \frac{\mathbf{A}^{1/2}}{\text{tr}(\mathbf{A}^{1/2})}, \quad L^* = 1 + \frac{\{\text{tr}(\mathbf{A}^{1/2})\}^2}{B}; \quad (3)$$

(ii) with matched means, the ratio of the optimal spread term to the representative-shaped one at the same budget is

$$\frac{L^* - 1}{L_{\text{rep}} - 1} = \frac{\{\text{tr}(\boldsymbol{\Sigma}_T^{1/2})\}^2}{p \text{tr}(\boldsymbol{\Sigma}_T)} \leq 1, \quad (4)$$

with equality if and only if $\boldsymbol{\Sigma}_T$ is a multiple of the identity; and (iii) $L^* \downarrow 1$ as $B \rightarrow \infty$. The proof, including the strict-convexity and constraint-qualification steps, is given in Appendix B.

Part (i) says that spread should follow the square root of the target’s covariance, not the covariance itself. Relative to the target’s own shape, the optimal source therefore flattens the allocation, granting proportionally more spread to the target’s narrow directions. Intuitively, the leverage cost of source narrowness in a direction is the reciprocal of the source variance there, so directions the target barely occupies still need protective spread; the square root is exactly the allocation that equalizes these marginal costs. The anisotropic scenario mentioned in Section 2.1, in which a source narrow in one direction reached leverage 11.25, is precisely the failure this rule repairs. Figure 3(a) makes the geometry concrete for a two-moderator target with standard-deviation ratio 4: the square-root source is visibly rounder than the representative one, and at a matched budget its leverage is lower.

Part (ii) is a verdict on representativeness inside the trusted model. The representative shape is optimal exactly when the target is isotropic, and the efficiency ratio (4) quantifies the shortfall otherwise; Figure 3(b) traces the closed form for a two-moderator target as the standard-deviation ratio r grows, reaching 0.56 at $r = 16$. A companion fact sharpens the comparison and is worth recording: at a matched trace budget the representative and isotropic shapes both deliver spread term exactly p , hence leverage exactly q , for every target anisotropy, so the two natural default shapes cannot be separated on leverage grounds; the square-root shape is the unique improvement, and shapes that invert the target are punished rapidly. In the same two-moderator family, the anti-target shape reaches leverage 257 at $r = 16$, against 2.12 for the square-root rule (Figure 3(c)).

Part (iii) is the warning. As the budget grows the optimal leverage falls to its absolute floor of one, so within the trusted model no finite budget is ever preferred

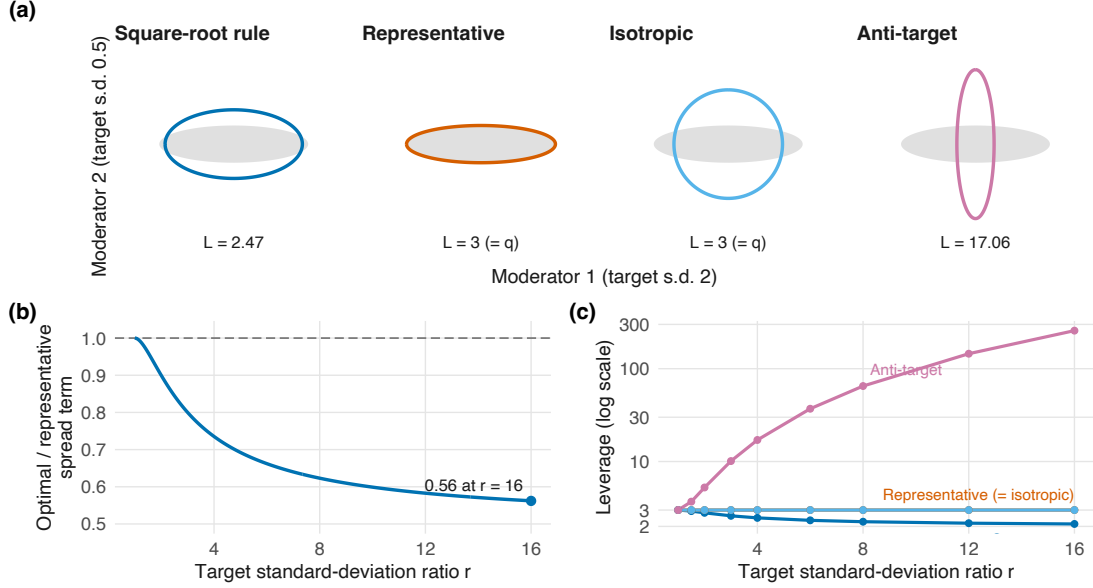


FIGURE 3. The square-root allocation (Theorem 1; all values exact). (a) Equal-budget source shapes for a two-moderator target with standard-deviation ratio $r = 4$ (target ellipse shaded): the square-root source is rounder than the representative one, and the anti-target shape concentrates spread where the target is narrow; leverage values are printed beneath each shape. (b) The efficiency ratio (4) against target anisotropy: the representative shape is optimal only at $r = 1$. (c) Leverage of the four shapes at the same trace budget (log scale): representative and isotropic coincide at $q = 3$ for every r , and the anti-target shape deteriorates rapidly.

to a larger one. Compact covariate supports, feasibility, and recruitment cost all bound spread in practice, but it seems to us unsatisfactory for the statistical calculus to rely on external constraints to rein in its own recommendation. Section 6 shows that model doubt, expressed as bounded curvature, supplies the counterweight from within.

5. Selecting Sites: A Convex Program with an Exact Support Certificate

Designers rarely choose a covariance freely. More often they assemble the sample from sites, districts, or strata that come with covariate profiles of their own, and the design variable is the vector of enrollment shares. Let candidate sites $k = 1, \dots, K$ have augmented second moments $\mathbf{M}_k$, and let a design be a weight vector $\mathbf{w}$ on the simplex, giving the mixture moment $\mathbf{M}_S(\mathbf{w}) = \sum_k w_k \mathbf{M}_k$. The mixture moment

prices mean shift and spread jointly, which is exactly where the scope limitation of Proposition 1 is resolved: a site program never trades means against spread by accident, because both live inside the same matrix.

Proposition 2 (Site selection). The map $\mathbf{w} \mapsto L(\mathbf{w}) = \text{tr}\{\mathbf{M}_S(\mathbf{w})^{-1}\mathbf{M}_T\}$ is convex on the simplex wherever $\mathbf{M}_S(\mathbf{w}) \succ \mathbf{0}$. A feasible $\mathbf{w}^*$ is optimal if and only if

$$\max_k \text{tr}\{\mathbf{M}_S(\mathbf{w}^*)^{-1}\mathbf{M}_T\mathbf{M}_S(\mathbf{w}^*)^{-1}\mathbf{M}_k\} = L(\mathbf{w}^*), \quad (5)$$

which is the leverage form of the classical equivalence condition (Kiefer, 1974; Kiefer & Wolfowitz, 1960); and some optimum puts positive weight on at most $q(q+1)/2$ sites, because the candidate moments share a unit $(1,1)$ entry and therefore lie in an affine hyperplane of the symmetric-matrix space, where Carathéodory’s theorem applies. Proofs, which specialize standard optimal-design arguments (Pukelsheim, 1993), are given in Appendix C.

Proposition 2 makes the site decision computable in a strong sense. Because the program is convex, it can be solved reliably; our implementation runs pairwise Frank–Wolfe iterations followed by an active-set polish, and condition (5) then certifies the solution, with a relative gap below 2×10^{-8} in the example below. The site sensitivities on the left of (5) report each site’s marginal value at the current design, so the solution is also auditable: a recruitment team can see which candidate sites the certificate says are worth adding. Moreover, any feasible weight vector can be converted, by an exact Carathéodory reduction, into a design on at most $q(q+1)/2$ sites with the *same* mixture moment matrix, so the small support is itself a certificate rather than a numerical-threshold judgment. We describe the reduction algorithm, together with a forced stress test in which thirty strictly positive weights are reduced to six while the moment matrix is preserved to machine precision, in Appendix C; reporting support by thresholding small weights instead is unreliable, and in this pool it would misreport the answer.

Figure 4 shows the worked example: a pool of thirty candidate sites with two moderators, each site carrying its own mean and covariance. The certified optimal design enrolls four sites (the representation bound allows six) at leverage 1.51. Equal enrollment across all thirty sites gives 2.27, the best single site gives 2.63, and the intuitive heuristic of enrolling the five sites whose mean profiles are nearest the target’s delivers 3.67, the worst of the four designs evaluated and 2.4 times the optimum. The heuristic fails for the reason the decomposition (2) makes plain: proximity in means says nothing about spread, and a cluster of near-target sites is typically an

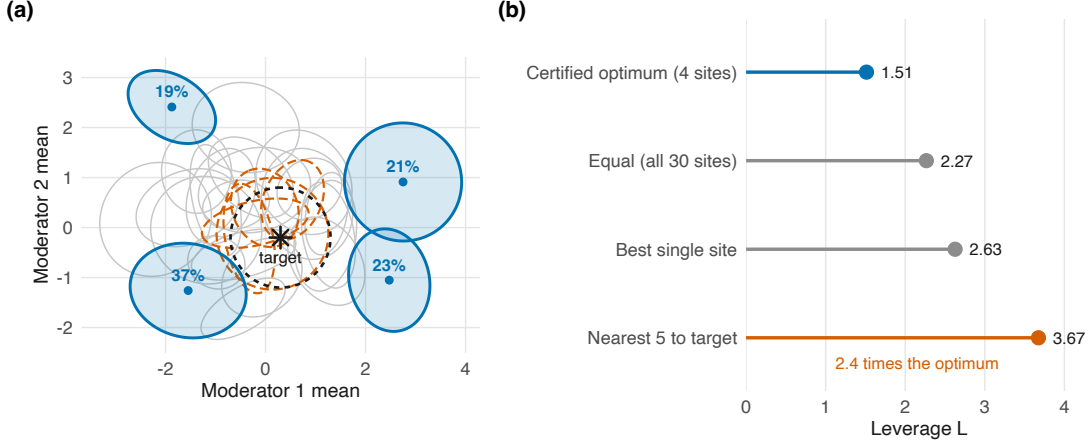


FIGURE 4. Site selection over a pool of $K = 30$ candidate sites with $p = 2$ moderators (Proposition 2). (a) Candidate sites drawn as covariance ellipses at their mean profiles; the four sites carrying the certified optimal weights are highlighted with their enrollment shares, the star and dashed circle mark the target mean and covariance, and the five sites nearest the target in mean are outlined for comparison. (b) Leverage of the four evaluated designs; the nearest-five heuristic is the worst at 2.4 times the certified optimum. The equivalence gap of the polished solution is below 2×10^{-8} ; the exact support-reduction certificate is described in Appendix C.

underdispersed source whose dispersion term explodes. In the map, the certificate instead spends its weight on sites that surround the target from moderately far away, because distant sites earn their enrollment by contributing variance in directions the near-target cluster lacks. We would caution against reading the specific numbers as more than an illustration for this pool, but the mechanism they illustrate is general.

6. Curvature and the Limits of Deliberate Spread

Everything so far treats the working model as exact, and Section 4 showed where that trust leads: leverage rewards unbounded spread. In this section we bound the model’s possible curvature and ask what allocation the bound recommends. We work first in one moderator dimension, where every object is available in closed form, and then in general dimension, where the answer changes shape in an instructive way. Throughout, we are explicit that the resulting objective is a planning criterion: it captures the design-dependent part of the worst-case bias exactly, and Section 6.3 prices what it leaves out.

6.1. One moderator. Let the source be symmetric with mean zero and spread $u = s_g^2$, let the target have mean zero, unit variance, and finite fourth moment, and let the true arm-wise effect functions include quadratic terms whose contrast coefficient is at most κ in absolute value. For a source of this kind the population least-squares projection leaves the fitted slope uncontaminated, and the worst-case squared extrapolation bias that the curvature bound permits, integrated over the target, has a design-dependent part proportional to $(u - 1)^2$ (Appendix D). Adding the first-order estimation component with $a = 2\sigma^2/n$ gives the design criterion

$$g(u) = a \left(1 + \frac{1}{u}\right) + \kappa^2(u - 1)^2. \quad (6)$$

Theorem 2 (Interior optimum under curvature; scalar). For $a > 0$ and $\kappa > 0$:
(i) g has a unique minimizer $u^* \in (1, \infty)$, characterized by $2\kappa^2 u^2(u - 1) = a$; and
(ii) u^* is strictly decreasing in n and in κ , with $u^* \rightarrow 1$ as $n \rightarrow \infty$ or $\kappa \rightarrow \infty$ and $u^* \rightarrow \infty$ as $\kappa \rightarrow 0$. The derivation, including the closed-form worst-case bias, is given in Appendix D.

The curvature bound thus restores an interior optimum, and the comparative statics say when deliberate overdispersion is worth having. At $\kappa = 0.15$, for instance, the optimal spread falls from 1.74 at $n = 20$ per arm to 1.16 at $n = 200$ and 1.04 at $n = 1,000$ (Figure 5(a, b)). Overdispersion is accordingly a tactic for small samples under a trusted functional form, and it fades as either the sample grows or the model is trusted less. We find this reading of the comparative statics more useful than the fixed prescription, because κ is not estimable from the design and must be elicited; what the theorem supplies is how the recommendation responds when the elicited bound moves.

6.2. The multivariate criterion and the target's shape. In several dimensions the trusted calculus of Theorem 1 tilts the allocation toward the square-root shape, so the question becomes whether the tilt survives model doubt. Work in target-whitened coordinates, so that the centered target has covariance $\mathbf{I}_p$ and matched means; assume the source law is centrally symmetric about zero (more generally, its third-moment tensor must vanish, a condition the projection argument requires). Let the true arm-wise effect functions include quadratic contamination $\mathbf{x}^\top \mathbf{C}_a \mathbf{x}$ whose *contrast* matrix $\mathbf{C} = \mathbf{C}_1 - \mathbf{C}_0$ satisfies $\|\mathbf{C}\| \leq \kappa$ in spectral norm. For fixed $\mathbf{C}$, the design-dependent part of the target-integrated squared contrast bias equals $\{\text{tr}(\mathbf{C}(\mathbf{I} - \boldsymbol{\Sigma}))\}^2$ (Appendix D), and its supremum over the spectral ball is $\kappa^2 \|\boldsymbol{\Sigma} - \mathbf{I}\|_*^2$, a squared

nuclear norm. We therefore study the planning criterion

$$F(\boldsymbol{\Sigma}) = a \{1 + \text{tr}(\boldsymbol{\Sigma}^{-1})\} + \kappa^2 \|\boldsymbol{\Sigma} - \mathbf{I}\|_*^2, \quad a = \frac{2\sigma^2}{n}. \quad (7)$$

Theorem 3 (Criterion-optimal target shape; multivariate). For every $a > 0$ and $\kappa > 0$, the criterion F in (7) has a unique minimizer over $\boldsymbol{\Sigma} \succ \mathbf{0}$, and it is isotropic: $\boldsymbol{\Sigma}^* = u^* \mathbf{I}_p$, where $u^* \in (1, \infty)$ is the unique root of

$$2p\kappa^2 u^2(u - 1) = a. \quad (8)$$

Moreover u^* is strictly decreasing in n , in κ , and in p , with $u^* \rightarrow 1$ in each limit. In original coordinates, the F -optimal source covariance has the target covariance shape inflated by u^* . For a Gaussian target, one contrast matrix can simultaneously maximize the design-dependent term and the design-free variance term, including when $\boldsymbol{\Sigma}$ has eigenvalue one; the latter contributes the constant $2p\kappa^2$, so it does not change this optimizer. For a non-Gaussian target, F deliberately controls only the design-dependent contrast-bias term. Even in the Gaussian case this is not complete finite-sample total-risk minimax: bounding $\mathbf{C}_1 - \mathbf{C}_0$ leaves common arm curvature unrestricted, while sandwich variance depends on $\mathbf{C}_1$ and $\mathbf{C}_0$ separately. The proof and the joint-maximizer construction are given in Appendix D.

Theorem 3 is the central criterion result of the article. Within F , the nuclear penalty removes the square-root tilt of Theorem 1 entirely and restores the target’s own shape; all that survives of deliberate heterogeneity is a single overdispersion factor, and that factor tends to one. The dimension effect is perhaps the most useful of the comparative statics, and it arises because the nuclear mismatch accumulates across directions inside F rather than through any budget: at $n = 50$ and $\kappa = 0.15$, the optimal inflation u^* is 1.43 for one moderator but 1.27, 1.20, 1.14, and 1.08 for $p = 2, 3, 5, 10$ (Figure 5(c)). With many moderators, in other words, the criterion leaves very little room for deliberate overdispersion. The qualitative compromise between variance and extrapolation bias, and the return to isotropic or rotatable geometry under model protection, are classical themes (Box & Draper, 1959; Steinberg, 1985; Wiens, 1992, 1994); what Theorem 3 adds is the target-whitened USTE formulation, the spectral contrast-curvature ball with its squared-nuclear-norm form, and the closed-form n -, κ -, and p -statics. It does not say that $u^* > 1$ minimizes complete total risk, and the next subsection explains why we insist on the distinction.

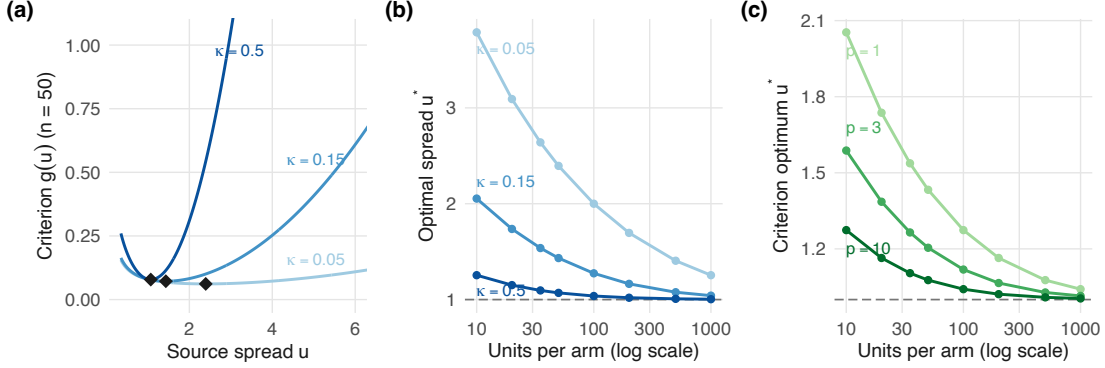


FIGURE 5. Curvature bounds restore an interior optimum and then erode it (closed forms). (a) The scalar criterion (6) at $n = 50$ per arm for three curvature bounds, with interior optima marked. (b) The optimal spread u^* against per-arm sample size for the same bounds; every path decreases toward the representative value $u = 1$ (Theorem 2). (c) The multivariate criterion optimum of Theorem 3 for $p \in \{1, 3, 10\}$ moderators at $\kappa = 0.15$; larger dimension approaches representativeness faster within F .

6.3. The complete first-order risk. A planning criterion earns trust by being explicit about its omissions, so we now compute the complete first-order total risk in the one case where every term is available in closed form, and compare. Fix the Gaussian scale family $x \sim \mathcal{N}(0, u)$ with contrast curvature κ split evenly across arms ($c_1 = \kappa/2 = -c_0$), a unit-variance Gaussian target, and residual scale σ .

Proposition 3 (Complete first-order total risk; Gaussian scale family). The total target risk of the OLS contrast satisfies

$$R(u) = \kappa^2 \{(u - 1)^2 + \text{Var}_T(x^2)\} + \frac{2\sigma^2(1 + 1/u) + 3\kappa^2 u(3 - u)}{n} + O(n^{-2}), \quad (9)$$

in which the curvature part of the $1/n$ term decomposes as a sandwich-variance contribution $\kappa^2(u^2 + 5u)/n$ and a projection-drift cross term $-4\kappa^2 u(u - 1)/n$, the latter arising because the realized projection intercept is biased by $-2\kappa u/n$ relative to the population projection. The derivation is given in Appendix D, together with a scale-family proposition covering non-Gaussian sources: for $x = \sqrt{u}Z$ with fixed symmetric Z of finite sixth moment, the per-arm sandwich addition is nondecreasing in u , strictly increasing exactly when $c_a \neq 0$ and $\Pr(Z^2 \neq 1) > 0$, and identically zero for Rademacher Z .

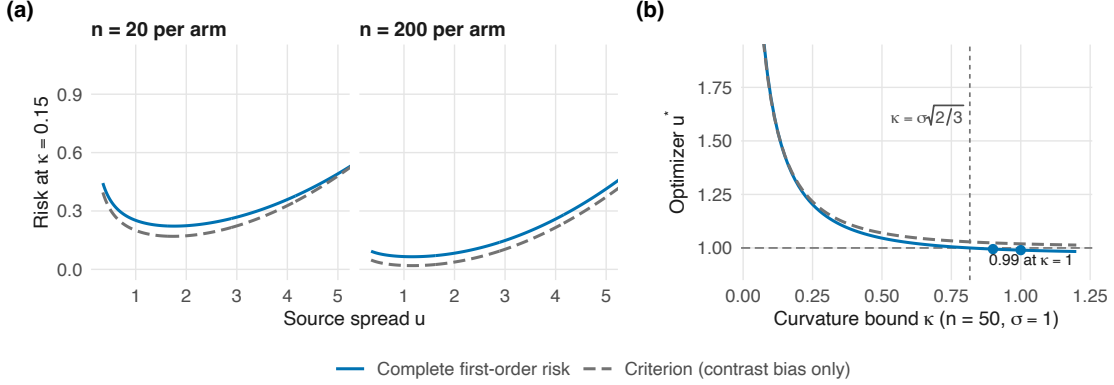


FIGURE 6. The criterion and the complete first-order risk (Proposition 3; closed forms, Monte Carlo agreement in Appendix E). (a) At the mild default $\kappa = 0.15$ the two layers nearly coincide at $n = 20$ and $n = 200$ per arm, and both prefer modest overdispersion. (b) Optimizer paths in κ at $n = 50$, $\sigma = 1$: the criterion optimum stays above the representative spread, whereas the complete-risk optimum crosses below it at $\kappa = \sigma\sqrt{2/3}$ (dotted line), the threshold implied by the derivative of (9) at $u = 1$.

Three remarks connect the proposition to practice. First, $\text{Var}_T(x^2)$ does not involve u : it raises the level of the risk but cannot move the optimizer, which is why the criterion (6) could omit it harmlessly. Second, the projection-drift term is genuinely first order, and dropping it is detectable: across $n \in \{50, 200, 800\}$ and $u \in \{1, 2, 4\}$, the complete formula (9) matches a 30,000-replication Monte Carlo within 0.13 percent in every cell, whereas the sandwich-only version misses by up to 8.7 percent at $n = 50$ with a discrepancy decaying at the $1/n$ rate that identifies a missing first-order term. Third, and most importantly for design advice, the agreement between criterion and complete risk that Figure 6(a) displays at $\kappa = 0.15$ is specific to mild curvature. For $n > 3$ the displayed first-order risk is strictly convex with derivative $(3\kappa^2 - 2\sigma^2)/n$ at $u = 1$, so its unique optimizer falls *below* one exactly when $\kappa^2 > 2\sigma^2/3$. At $n = 50$ and $\sigma = 1$, the complete-risk optima are 0.9946 at $\kappa = 0.9$ and 0.9898 at $\kappa = 1$, while the stated criterion still recommends 1.0236 and 1.0193 (Figure 6(b)). Under strong curvature, that is, the complete first-order accounting can prefer a source slightly *narrower* than the target, essentially because heavily curved arms make wide sources noisy as well as biased. We conclude that (6) and (7) are transparent planning devices for the contrast-bias component, and that a designer contemplating substantial curvature should run the complete-risk sensitivity before committing to overdispersion.

7. Noise, Finite Samples, and Exact Comparisons

The leverage calculus compares sources through their covariate moments alone. Real candidate sources differ in two further ways that a design comparison should not ignore: their residual noise, and the finite-sample behavior of their realized design matrices. This section prices both.

7.1. Noise-adjusted comparison. A leverage ranking of candidate sources becomes a risk ranking only after each source’s residual noise is priced in, and the honest comparison simply multiplies the two.

Corollary 1 (Noise-adjusted comparison). At first order, with a common per-arm size and a common floor, source S_1 delivers lower target risk than S_2 if and only if $\bar{\sigma}^2(S_1) L_{T|S_1} < \bar{\sigma}^2(S_2) L_{T|S_2}$, where $\bar{\sigma}^2$ is the summed arm residual variance. With a linear noise profile $\sigma^2(u) = 1 + g(u - 1)$ in one dimension, the objective $\sigma^2(u)\{1 + 1/u\}$ is minimized on the interior at

$$u_{\text{noise}}^* = \sqrt{\frac{1-g}{g}}, \quad 0 < g < 1, \quad (10)$$

and the domain restriction matters: at $g = 0$ the objective decreases without bound, as $g \uparrow 1$ the optimum tends to zero spread, and outside the range where the affine profile is positive the formula does not apply. The derivation is given in Appendix D.

The first statement is the honest form of the leverage comparison when sites differ in more than covariates: a source with half the leverage still loses whenever its residual variance is more than twice as large, which is the reversal example of the companion article. The profile rule (10) then describes how the optimal spread responds when noise itself varies with the covariates being recruited. Its behavior is not monotone advice for or against heterogeneity. The optimum sits above the representative spread for gradients $g < 1/2$ but below it for $g > 1/2$; at $g = 0.8$, for instance, the rule recommends $u = 0.50$, a source deliberately narrower than the target (Figure 7). A designer therefore needs local knowledge of how residual noise varies across the recruitable range, and leverage supplies only half of the comparison.

7.2. Finite-sample behavior. The first-order calculus also sees the source only through its population moments, and two finite-sample facts qualify that view. First, among sources with identical covariance, lighter tails inflate less. The cleanest version of the fact is exact: when n is divisible by 2^p , the balanced two-point factorial design

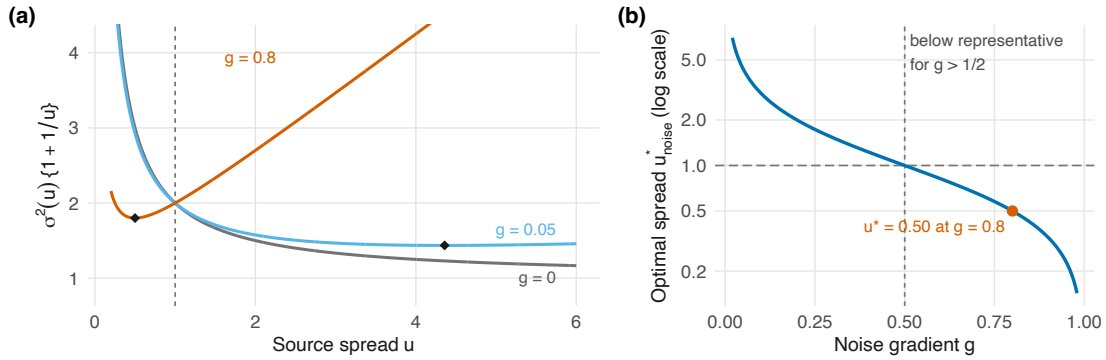


FIGURE 7. The noise-leverage interaction for one moderator at first order (Corollary 1). (a) The objective $\sigma^2(u)\{1 + 1/u\}$ under noise profiles $\sigma^2(u) = 1 + g(u - 1)$ with gradients $g \in \{0, 0.05, 0.8\}$; interior optima are marked, and the flat-noise profile ($g = 0$) decreases without bound. (b) The profile rule (10) across its domain: the optimal spread crosses the representative value at $g = 1/2$ and falls below it for steeper gradients.

has augmented cross-product exactly $n\mathbf{I}_q$ and achieves the first-order factor with no inflation at all, whereas independently drawn designs inflate through realized-moment variability. At $n = 20$ and $p = 2$, the mean realized factor is 0.150 for the balanced factorial against a first-order value of 0.150, and rises to 0.169, 0.174, and 0.181 for i.i.d. two-point, uniform, and Gaussian sources respectively (Figure 8(a)). The classical affinity of designed experiments for extreme-point support thus survives in this problem as a tail-weight preference, although the same geometry is of course maximally exposed to the curvature costs of Section 6; the sampling protocols behind the comparison, including the redraw rule that guards the smallest sample size, are stated in Appendix E. Second, the inflation depends on spread. Using the companion’s exact Gaussian-design formula, the exact-to-first-order ratio at $n = 20$ is 1.21 at $u = 1$ but 1.17 at $u = 4$ (Figure 8(b)), so first-order comparisons made at small sizes are biased against exactly the overdispersed designs that Section 6 might otherwise justify. For final comparisons at realistic sizes we therefore recommend the exact expressions rather than (1).

8. Discussion

We have asked how an experimental sample should be designed when the goal is unit-specific treatment-effect prediction for a stated target population, and the organizing object throughout has been the relative leverage factor. Under the trusted

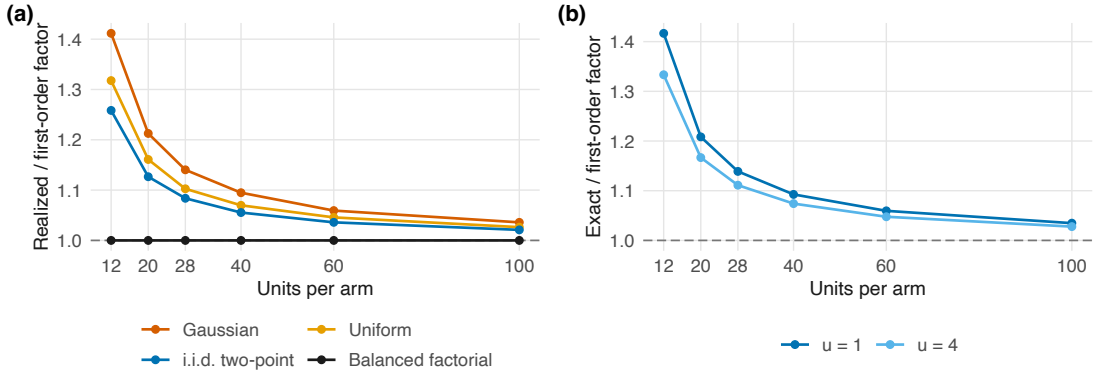


FIGURE 8. Finite-sample qualifications for $p = 2$ (protocols in Appendix E). (a) Mean realized-to-first-order factor for four design protocols with identical covariance: the balanced factorial, exact whenever n is divisible by 2^p , sits at one; independently drawn designs inflate in order of tail weight. (b) The exact-to-first-order ratio for Gaussian sources at two spreads: the first-order approximation is more optimistic for narrow sources than for wide ones at small n .

working model the answers are constructive. Matching the target’s covariate means is always advantageous at fixed dispersion; a spread budget is best allocated by the matrix square root of the target covariance, which improves on the representative shape except for isotropic targets; and site selection is a convex program whose solution is certified by an equivalence condition and representable on a provably small set of sites. In our thirty-site pool the certified optimum used four sites, and the popular heuristic of enrolling the sites nearest the target was the worst design evaluated, which we take as a caution that intuitive similarity is a poor proxy for prediction value.

The same trusted calculus, however, rewards unbounded spread, and we have treated that as a defect to be repaired inside the theory rather than left to outside judgment. Bounding the curvature of the treatment-effect contrast repairs it: an interior optimum returns, and in several dimensions the criterion-optimal covariance abandons the square-root tilt for the target’s own shape, inflated by a factor that fades with sample size, with the curvature bound, and with the number of moderators. Under this criterion, that is, a designer who doubts the model is led back toward representative sampling, and deliberate heterogeneity survives only as a modest and shrinking inflation. At the same time the criterion is only a criterion. It controls the

design-dependent worst-case contrast bias, not the complete risk; the complete first-order accounting of Proposition 3 can prefer mild underdispersion when curvature is strong; noise gradients across the recruitable range can push the optimum in either direction; and finite-sample inflation penalizes narrow sources at small sizes. None of these refinements changes the qualitative story, but each can change a close decision, so we have quantified each rather than left it as a caveat.

For practice, the results are best used in the order a design conversation would encounter them, the order in which the sections developed them and in which Figure 1 arranges them. The target’s covariate means should be matched at whatever dispersion is contemplated, with joint mean-and-spread choices routed through the site program rather than made informally. The curvature bound κ should then be elicited and stated, not defaulted, because every recommendation in Section 6 is indexed by it, and whenever a contemplated design departs materially from the target’s shape, the complete-risk sensitivity of Proposition 3 should be run alongside the criterion, with overdispersion adopted only when both layers agree that it helps. The square-root tilt of Theorem 1 remains available wherever the linear working model is genuinely trusted. When the sample is assembled from sites, the convex program of Proposition 2 should replace similarity-based enrollment, and in our view it is the component most likely to be useful in routine practice, since it substitutes a certificate for an informal judgment. Finalists should then be compared on noise-adjusted risk with the exact finite-sample expressions of Section 7 evaluated at the actual sample size.

The limitations trace the article’s boundaries, and Appendix E records, claim by claim, what the accompanying verification suite supports and what it does not. The criterion F is worst case only for the design-dependent component induced by bounded quadratic contrast curvature; the projection argument requires vanishing source third moments, and the Gaussian-target assumption is needed for the joint worst-case interpretation. Departures outside that class, including shifts in higher moments or in residual structure, are not covered, and the transport-based ambiguity sets of Bouyamourn (2025) are complementary on exactly that front. The complete-risk expansion is proved for a Gaussian scale family with an even arm-curvature split, and without a separate bound on the individual arm curvatures no finite total-risk supremum exists, which is the structural reason we decline to call any design minimax. The site model omits participation behavior, richer cost structures, and within-site clustering (Litwok et al., 2023). An empirical application of the full sequence to a

school-level evaluation, from the realized leverage decomposition through the site program to the criterion-optimal covariance for an actual state sample against a national frame, is left to future work. The residual floor and its non-identified correlation remain exactly as the companion left them: no choice of source relaxes that floor, and the present article should be read as optimizing the estimation component that design can actually reach. Analysis-side tuning of the fitted model is the companion’s subject (Lee, 2026); the two articles meet at the leverage matrix, whose trace prices the design burden studied here and whose effective dimension governs the tuning guarantees studied there.

Reproducibility. All numerical claims in this article, including every figure, table, and in-text value, are generated by a documented verification and simulation suite, using closed-form computations, deterministic quadrature, and Monte Carlo with fixed seeds; no Markov chain Monte Carlo is used. Every verification block carries an evidence-type label (analytic proof, deterministic quadrature, independent numerical derivation, Monte Carlo with stated tolerance, regression check, or descriptive report), and the map from each formal statement to its blocks appears in Appendix E, together with the seeds and simulation settings. The suite (R scripts 00_utils.R, 01_verification.R, 02_simulation.R, and 03_figures.R, with a README) and the verification report from the shipped run accompany this article as arXiv ancillary files.

Data availability. This article analyzes no empirical data. Every reported quantity is either a closed-form consequence of the stated design model or the output of the simulation described in Appendix E, which generates its own inputs from the documented seeds.

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Online Supplement

Target-Weighted Experimental Sample Design for Predicting Unit-Specific Treatment Effects

JoonHo Lee

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Appendix A. Leverage Decomposition and Mean Matching

Lemma 1 (restated). With $\mathbf{M}_P = \begin{pmatrix} 1 & \boldsymbol{\mu}_P^\top \\ \boldsymbol{\mu}_P & \boldsymbol{\Sigma}_P + \boldsymbol{\mu}_P \boldsymbol{\mu}_P^\top \end{pmatrix}$ for $P \in \{S, T\}$ and $\boldsymbol{\Sigma}_S \succ \mathbf{0}$,

$$\text{tr}(\mathbf{M}_S^{-1} \mathbf{M}_T) = 1 + (\boldsymbol{\mu}_T - \boldsymbol{\mu}_S)^\top \boldsymbol{\Sigma}_S^{-1} (\boldsymbol{\mu}_T - \boldsymbol{\mu}_S) + \text{tr}(\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\Sigma}_T).$$

Proof. The Schur complement of the $(1, 1)$ entry of $\mathbf{M}_S$ is $\boldsymbol{\Sigma}_S$, so

$$\mathbf{M}_S^{-1} = \begin{pmatrix} 1 + \boldsymbol{\mu}_S^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S & -\boldsymbol{\mu}_S^\top \boldsymbol{\Sigma}_S^{-1} \\ -\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S & \boldsymbol{\Sigma}_S^{-1} \end{pmatrix}. \quad (\text{A.1})$$

Multiplying by $\mathbf{M}_T$ in block form, the two diagonal blocks of the product are

$$(1 + \boldsymbol{\mu}_S^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S) - \boldsymbol{\mu}_S^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_T \quad \text{and} \quad -\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S \boldsymbol{\mu}_T^\top + \boldsymbol{\Sigma}_S^{-1} (\boldsymbol{\Sigma}_T + \boldsymbol{\mu}_T \boldsymbol{\mu}_T^\top),$$

so the trace is

$$1 + \boldsymbol{\mu}_S^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S - \boldsymbol{\mu}_S^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_T - \boldsymbol{\mu}_T^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S + \text{tr}(\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\Sigma}_T) + \boldsymbol{\mu}_T^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_T,$$

using $\text{tr}(\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_T \boldsymbol{\mu}_T^\top) = \boldsymbol{\mu}_T^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_T$ and $\text{tr}(\boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S \boldsymbol{\mu}_T^\top) = \boldsymbol{\mu}_T^\top \boldsymbol{\Sigma}_S^{-1} \boldsymbol{\mu}_S$. Completing the square collects the four mean terms into $(\boldsymbol{\mu}_T - \boldsymbol{\mu}_S)^\top \boldsymbol{\Sigma}_S^{-1} (\boldsymbol{\mu}_T - \boldsymbol{\mu}_S)$, which is the claim. The decomposition is verified numerically over fifty random configurations with p between two and five (block V1; maximum absolute discrepancy 2.6×10^{-13}). $\square$

Proposition 1 (restated; fixed covariance). *For fixed $\boldsymbol{\Sigma}_S \succ \mathbf{0}$, the map $\boldsymbol{\mu}_S \mapsto L_{T|S}$ is uniquely minimized at $\boldsymbol{\mu}_S = \boldsymbol{\mu}_T$.*

Proof. Only the middle term of the decomposition involves $\boldsymbol{\mu}_S$; it is a positive definite quadratic form in $\boldsymbol{\mu}_T - \boldsymbol{\mu}_S$, strictly positive unless the difference vanishes. $\square$

With $\boldsymbol{\delta} = \boldsymbol{\mu}_T - \boldsymbol{\mu}_S$ fixed, the decomposition rewrites as $1 + \text{tr}\{\boldsymbol{\Sigma}_S^{-1}(\boldsymbol{\Sigma}_T + \boldsymbol{\delta} \boldsymbol{\delta}^\top)\}$, so Appendix B applies with the weight $\mathbf{A} = \boldsymbol{\Sigma}_T + \boldsymbol{\delta} \boldsymbol{\delta}^\top$. The main text states the fixed-covariance scope explicitly: in site problems the weights move means and covariances jointly, and the joint object is the mixture moment of Appendix C.

Appendix B. Proof of Theorem 1 (Square-Root Allocation)

Theorem 1 (restated). *For $\mathbf{A} \succ \mathbf{0}$ and $B > 0$, the problem $\min_{\boldsymbol{\Sigma} \succ \mathbf{0}} f(\boldsymbol{\Sigma}) = \text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{A})$ subject to $\text{tr}(\boldsymbol{\Sigma}) \leq B$ has the unique solution $\boldsymbol{\Sigma}^* = B \mathbf{A}^{1/2} / \text{tr}(\mathbf{A}^{1/2})$ with value $\{\text{tr}(\mathbf{A}^{1/2})\}^2 / B$; the efficiency ratio against the representative shape is $\{\text{tr}(\boldsymbol{\Sigma}_T^{1/2})\}^2 / \{p \text{tr}(\boldsymbol{\Sigma}_T)\} \leq 1$ at matched means, with equality iff $\boldsymbol{\Sigma}_T \propto \mathbf{I}$; and the value tends to zero as $B \rightarrow \infty$.*

Proof. Strict convexity. For symmetric $\mathbf{H} \neq \mathbf{0}$ and $\boldsymbol{\Sigma} \succ \mathbf{0}$, differentiate twice along $\boldsymbol{\Sigma} + t\mathbf{H}$:

$$\left. \frac{d^2}{dt^2} f(\boldsymbol{\Sigma} + t\mathbf{H}) \right|_{t=0} = 2 \text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{H} \boldsymbol{\Sigma}^{-1} \mathbf{H} \boldsymbol{\Sigma}^{-1} \mathbf{A}) = 2 \text{tr}(\mathbf{P} \mathbf{K}^2),$$

where $\mathbf{K} = \boldsymbol{\Sigma}^{-1/2} \mathbf{H} \boldsymbol{\Sigma}^{-1/2}$ and $\mathbf{P} = \boldsymbol{\Sigma}^{-1/2} \mathbf{A} \boldsymbol{\Sigma}^{-1/2} \succ \mathbf{0}$. Since $\mathbf{K}^2 \succeq \mathbf{0}$ with $\mathbf{K}^2 \neq \mathbf{0}$, and $\mathbf{P} \succ \mathbf{0}$, the trace is strictly positive: write $\text{tr}(\mathbf{P} \mathbf{K}^2) = \text{tr}\{(\mathbf{P}^{1/2} \mathbf{K})(\mathbf{P}^{1/2} \mathbf{K})^\top\} > 0$. So f is strictly convex on the positive definite cone.

The constraint binds and Slater holds. For any feasible Σ with $\text{tr}(\Sigma) < B$, scaling by $c = B/\text{tr}(\Sigma) > 1$ stays feasible and reduces the objective by the factor $1/c$; hence every minimizer satisfies $\text{tr}(\Sigma) = B$. The interior point $(B/(2p))\mathbf{I}$ is positive definite and has trace $B/2 < B$, so it is strictly feasible for the inequality form. Slater's condition holds, and the KKT conditions are necessary and sufficient for the convex problem.

Stationarity. With multiplier $\lambda \geq 0$ for the trace constraint, the stationarity condition is $-\Sigma^{-1}\mathbf{A}\Sigma^{-1} + \lambda\mathbf{I} = \mathbf{0}$, that is $\mathbf{A} = \lambda\Sigma^2$. Because $\mathbf{A} \succ \mathbf{0}$ we have $\lambda > 0$ and $\Sigma = \lambda^{-1/2}\mathbf{A}^{1/2}$, where $\mathbf{A}^{1/2}$ is the unique positive definite square root; no commutation needs to be assumed, because the equation itself forces Σ to be the positive root of $\mathbf{A}/\lambda$. The binding trace pins $\lambda^{-1/2}\text{tr}(\mathbf{A}^{1/2}) = B$, giving Σ^* and, upon substitution, $f(\Sigma^*) = \{\text{tr}(\mathbf{A}^{1/2})\}^2/B$. Uniqueness follows from strict convexity.

Efficiency and equality case. The representative shape $\Sigma_{\text{rep}} = B\mathbf{A}/\text{tr}(\mathbf{A})$ gives $f = \text{tr}(\mathbf{A})p/B$ when $\mathbf{A} = \Sigma_T$ (matched means), so the ratio of spread terms is $\{\sum_j \sqrt{a_j}\}^2/\{p \sum_j a_j\}$ over the eigenvalues a_j of Σ_T , which is at most one by Cauchy–Schwarz applied to $(\sqrt{a_1}, \dots, \sqrt{a_p})$ and $(1, \dots, 1)$, with equality iff all a_j are equal. The budget limit is immediate from the value formula. $\square$

Remark B.1. At a matched trace budget the representative and isotropic shapes both deliver spread term exactly p . For the representative shape,

$$\text{tr}\left\{\left(\frac{B\Sigma_T}{\text{tr}\Sigma_T}\right)^{-1}\Sigma_T\right\} = \frac{\text{tr}(\Sigma_T)p}{B} = p \quad \text{when } B = \text{tr}(\Sigma_T),$$

and the same holds for the isotropic shape. The anti-target shape $\Sigma \propto \Sigma_T^{-1}$ at the same budget reaches leverage 257.0 at $r = 16$ in the two-moderator family of the main text, against the square-root rule's 2.12 (table `tbl_shape_comparison.csv`). The closed form against a free Cholesky-parameterized optimization agrees to eight decimals (block V3a).

Appendix C. Site Selection: Proofs and the Exact Support Reduction

Proposition 2 (restated). *On $\{\mathbf{w} \in \Delta_{K-1} : \mathbf{M}(\mathbf{w}) = \sum_k w_k \mathbf{M}_k \succ \mathbf{0}\}$, the map $L(\mathbf{w}) = \text{tr}\{\mathbf{M}(\mathbf{w})^{-1}\mathbf{M}_T\}$ is convex; $\mathbf{w}^*$ is optimal iff $\max_k d_k(\mathbf{w}^*) = L(\mathbf{w}^*)$ with $d_k(\mathbf{w}) = \text{tr}\{\mathbf{M}(\mathbf{w})^{-1}\mathbf{M}_T\mathbf{M}(\mathbf{w})^{-1}\mathbf{M}_k\}$; and some optimum has support at most $q(q+1)/2$.*

Proof. Convexity. The map $\mathbf{M} \mapsto \text{tr}(\mathbf{M}^{-1}\mathbf{M}_T)$ is convex on the positive definite cone: for $\mathbf{M}_T \succeq \mathbf{0}$ it is a nonnegative linear functional of $\mathbf{M}^{-1}$, and $\mathbf{M} \mapsto \mathbf{M}^{-1}$ is operator convex, meaning $\{\alpha\mathbf{M}_1 + (1 - \alpha)\mathbf{M}_2\}^{-1} \preceq \alpha\mathbf{M}_1^{-1} + (1 - \alpha)\mathbf{M}_2^{-1}$; taking the trace against $\mathbf{M}_T$ preserves the inequality. Composition with the affine map $\mathbf{w} \mapsto \mathbf{M}(\mathbf{w})$ preserves convexity.

Equivalence. Using $d(\mathbf{M}^{-1}) = -\mathbf{M}^{-1}(d\mathbf{M})\mathbf{M}^{-1}$, the directional derivative of L at $\mathbf{w}$ toward the vertex $\mathbf{e}_k$ is

$$\left. \frac{d}{dt} L((1-t)\mathbf{w} + t\mathbf{e}_k) \right|_{t=0+} = L(\mathbf{w}) - d_k(\mathbf{w}).$$

For a convex function on the simplex, $\mathbf{w}^*$ is a minimizer iff no vertex direction is a descent direction, that is $d_k(\mathbf{w}^*) \leq L(\mathbf{w}^*)$ for all k ; and multiplying the identity $\sum_k w_k d_k(\mathbf{w}) = \text{tr}\{\mathbf{M}^{-1}\mathbf{M}_T\mathbf{M}^{-1}\mathbf{M}(\mathbf{w})\} = L(\mathbf{w})$ by the optimal weights shows every supported site attains equality. This is the leverage form of the general equivalence theory of Kiefer and Wolfowitz (1960) and Kiefer (1974).

Support bound. $L(\mathbf{w})$ depends on $\mathbf{w}$ only through $\mathbf{M}(\mathbf{w})$, which lies in the convex hull of $\{\mathbf{M}_k\}$ inside the d -dimensional space of symmetric $q \times q$ matrices, $d = q(q+1)/2$. Every candidate has $(1,1)$ entry equal to one, so the hull lies in an affine hyperplane of dimension at most $d-1$, and Carathéodory's theorem represents $\mathbf{M}(\mathbf{w}^*)$ as a convex combination of at most $(d-1) + 1 = d = q(q+1)/2$ candidates. Any such representation has the same leverage, because the objective is a function of the moment matrix alone. For $q = 3$ the bound is six, which block V4c tests directly. $\square$

C.1. The exact reduction algorithm. The solver (pairwise Frank–Wolfe with an active-set polish; relative equivalence gap 1.9×10^{-8} in the shipped run, block V4a) returns weights $\mathbf{w}$ whose support may exceed the bound or carry algorithmic remnants. The reduction takes any $\mathbf{w}$ and returns weights on at most $q(q+1)/2$ sites with the *same* moment matrix:

- (1) Let $\mathcal{A} = \{k : w_k > 0\}$ and stack the half-vectorized candidates $\text{vech}(\mathbf{M}_k)$, $k \in \mathcal{A}$, as columns of $\mathbf{V}_{\mathcal{A}}$ (d rows). Because every column has the same $(1,1)$ entry, any null vector $\mathbf{z}$ of $\mathbf{V}_{\mathcal{A}}$ automatically satisfies $\sum_k z_k = 0$.
- (2) While $|\mathcal{A}| > d$, compute the complete right-singular basis of $\mathbf{V}_{\mathcal{A}}$ and select a vector $\mathbf{z}$ from its numerical null-space basis. Requesting all $|\mathcal{A}|$ right singular vectors is essential when the matrix is wide; the default truncated basis contains no null vectors. Move $\mathbf{w}_{\mathcal{A}} \mapsto \mathbf{w}_{\mathcal{A}} + t\mathbf{z}$ with $t = \min_{z_k < 0} w_k / (-z_k)$ and set the minimizing coordinate exactly to zero. The common affine coordinate

gives $\sum_k z_k = 0$, while $\mathbf{V}_{\mathcal{A}}\mathbf{z} = \mathbf{0}$ preserves the moment matrix; nonnegativity and simplex mass are checked after every pivot.

- (3) Stop at $|\mathcal{A}| \leq d$ and report the support, minimum weight, simplex discrepancy, and maximum moment-matrix discrepancy.

Each pivot preserves $\mathbf{M}(\mathbf{w})$ to numerical precision, so the reduced design is a re-representation rather than a new approximate optimum. The polished optimum already has four sites. The mandatory regression therefore starts from uniform positive weights on all thirty sites: it reduces support $30 \rightarrow 6$, preserves simplex mass with nonnegative weights, and has maximum moment-entry discrepancy 4.4×10^{-16} (block V4f). This forced dense input prevents an identity function from passing merely because the optimizer was sparse before reduction.

Appendix D. Contrast-Curvature Criteria, the Complete Risk, and Noise

D.1. The scalar bias calculation and Theorem 2. Let the source be symmetric with mean zero and variance u , the target have mean zero, unit variance, and finite fourth moment, and let the arm-wise effect functions be $f_a(x) = \beta_{a0} + \beta_{a1}x + c_ax^2$ with contrast curvature $|c_1 - c_0| \leq \kappa$. By source symmetry, $\text{Cov}_S(x, x^2) = \mathbb{E}_S(x^3) = 0$, so the population projection of f_a onto $(1, x)$ leaves the slope uncontaminated and absorbs $c_a u$ into the intercept. The contrast bias function is therefore

$$b(x) = (c_1 - c_0)(u - x^2), \quad (\text{D.1})$$

and its target-integrated square is $(c_1 - c_0)^2 \{(u - 1)^2 + \text{Var}_T(x^2)\}$, using $\mathbb{E}_T(x^2) = 1$. The worst case over the class is attained at $|c_1 - c_0| = \kappa$. Combining the design-dependent part with the first-order estimation component $a(1 + 1/u)$, $a = 2\sigma^2/n$, gives the stated criterion $g(u) = a(1 + 1/u) + \kappa^2(u - 1)^2$; the term $\kappa^2 \text{Var}_T(x^2)$ is constant in u and is omitted from the criterion because it cannot move the optimizer.

Theorem 2 (restated; scalar). *For $a > 0$ and $\kappa > 0$: (i) g has a unique minimizer $u^* \in (1, \infty)$ solving $2\kappa^2 u^2(u - 1) = a$; (ii) u^* is strictly decreasing in n and κ , with $u^* \rightarrow 1$ as $n \rightarrow \infty$ or $\kappa \rightarrow \infty$, and $u^* \rightarrow \infty$ as $\kappa \rightarrow 0$.*

Proof. $g'(u) = -a/u^2 + 2\kappa^2(u - 1)$ is strictly increasing on $(0, \infty)$ (both summands increase), negative on $(0, 1]$, and tends to $+\infty$, so it has a unique root $u^* > 1$, which is the unique minimizer; rearranging $g'(u^*) = 0$ gives the cubic characterization. For (ii), apply the implicit function theorem to $F(u, a, \kappa) = 2\kappa^2 u^2(u - 1) - a$: $\partial F/\partial u = 2\kappa^2 u(3u - 2) > 0$ for $u > 1$, $\partial F/\partial a = -1 < 0$, and $\partial F/\partial \kappa = 4\kappa u^2(u - 1) > 0$ for $u > 1$, so $du^*/da > 0$ (hence decreasing in n) and $du^*/d\kappa < 0$. For the limits: as

$a \downarrow 0$ or $\kappa \uparrow \infty$ the cubic forces $u^2(u-1) \downarrow 0$ with $u > 1$, so $u^* \downarrow 1$; as $\kappa \downarrow 0$ it forces $u^2(u-1) \uparrow \infty$, so $u^* \uparrow \infty$. Blocks V5a through V5c verify the single sign change, the cubic residual (10^{-13}), and the monotone paths. $\square$

D.2. The multivariate criterion and Theorem 3. Work in target-whitened coordinates ($\Sigma_T = \mathbf{I}_p$, matched means) with source covariance $\Sigma \succ \mathbf{0}$ and contamination $\mathbf{x}^\top \mathbf{C}_a \mathbf{x}$, contrast matrix $\mathbf{C} = \mathbf{C}_1 - \mathbf{C}_0$ with $\|\mathbf{C}\| \leq \kappa$ (spectral norm). Assume that the centered source law has zero third-moment tensor, $\mathbb{E}_S(x_j x_k x_\ell) = 0$ for every j, k, ℓ ; central symmetry is a sufficient condition. This assumption is essential to the interpretation of F , because it makes the population projection leave the slopes uncontaminated and absorb $\mathbb{E}_S(\mathbf{x}^\top \mathbf{C}_a \mathbf{x}) = \text{tr}(\mathbf{C}_a \Sigma)$ into the intercepts. The contrast bias function is $b(\mathbf{x}) = \mathbf{x}^\top \mathbf{C} \mathbf{x} - \text{tr}(\mathbf{C} \Sigma)$, and its target-integrated square splits as

$$\mathbb{E}_T b(\mathbf{x})^2 = \text{Var}_T(\mathbf{x}^\top \mathbf{C} \mathbf{x}) + \{\text{tr}(\mathbf{C}(\mathbf{I} - \Sigma))\}^2, \quad (\text{D.2})$$

because $\mathbb{E}_T(\mathbf{x}^\top \mathbf{C} \mathbf{x}) = \text{tr}(\mathbf{C})$. The first term does not involve the design; the second is the design-dependent part. Its worst case over the spectral ball is

$$\sup_{\|\mathbf{C}\| \leq \kappa} \{\text{tr}(\mathbf{C}(\mathbf{I} - \Sigma))\}^2 = \kappa^2 \left(\sum_j |1 - \lambda_j(\Sigma)| \right)^2 = \kappa^2 \|\Sigma - \mathbf{I}\|_*^2.$$

To state attainment correctly when $\lambda_j(\Sigma) = 1$, choose $s_j = \text{sign}(1 - \lambda_j)$ for nonzero mismatch and choose either $s_j = -1$ or $s_j = 1$ on every zero-mismatch eigenspace. Then $\mathbf{C}^* = \kappa \sum_j s_j \mathbf{u}_j \mathbf{u}_j^\top$ attains the display and also satisfies $\text{tr}\{(\mathbf{C}^*)^2\} = p\kappa^2$. If the target is Gaussian, $\text{Var}_T(\mathbf{x}^\top \mathbf{C} \mathbf{x}) = 2\text{tr}(\mathbf{C}^2)$, so this saturated aligned matrix simultaneously maximizes both terms and

$$\sup_{\|\mathbf{C}\| \leq \kappa} \mathbb{E}_T b(\mathbf{x})^2 = 2p\kappa^2 + \kappa^2 \|\Sigma - \mathbf{I}\|_*^2.$$

The first term is constant in the design. For non-Gaussian targets the maximizing orientation of the design-free variance need not align, so F controls the worst-case design-dependent contrast-bias term rather than the joint worst case. Adding the estimation component $a\{1 + \text{tr}(\Sigma^{-1})\}$ gives $F(\Sigma)$. This variance–bias design tradition and the recovery of isotropic or rotatable geometry under model protection are classical (Box & Draper, 1959; Steinberg, 1985; Wiens, 1992, 1994); the result here is the target-whitened spectral-ball specialization. In either target case, a spectral bound on $\mathbf{C}_1 - \mathbf{C}_0$ alone does not bound the two arm-specific curvature variances, so F is not a complete finite-sample total-risk minimax criterion.

Theorem 3 (restated; criterion-optimal target shape). *F has the unique minimizer $\Sigma^* = u^* \mathbf{I}_p$ over $\Sigma \succ \mathbf{0}$, with $u^* \in (1, \infty)$ the unique root of $2p\kappa^2 u^2(u-1) = a$; and u^* is strictly decreasing in n , κ , and p , with limit one in each.*

Proof. Both terms of F are spectral functions of Σ : writing $\lambda = (\lambda_1, \dots, \lambda_p)$ for the eigenvalues,

$$F = a \left(1 + \sum_j \lambda_j^{-1} \right) + \kappa^2 \left(\sum_j |\lambda_j - 1| \right)^2 =: \Phi(\lambda).$$

Φ is convex on $(0, \infty)^p$: the first sum is strictly convex, and the second is the square of a nonnegative convex function, hence convex. Existence is not automatic on this open domain, so we record coercivity. If any $\lambda_j \downarrow 0$, the inverse term diverges; if $\max_j \lambda_j \rightarrow \infty$, then $\sum_j |\lambda_j - 1| \rightarrow \infty$ and the nuclear penalty diverges. Every sublevel set is therefore contained in some compact box $[\epsilon, M]^p$, and continuity gives attainment. As a function of the matrix, $\text{tr}(\Sigma^{-1})$ is strictly convex on the positive definite cone and $\|\Sigma - \mathbf{I}\|_*^2$ is convex, so F is strictly convex and the attained minimizer is unique. Φ is symmetric under permutations of λ ; a unique minimizer invariant under these symmetries must have all coordinates equal, say $\lambda_j \equiv u$, giving $\Sigma^* = u^* \mathbf{I}$. On the isotropic ray,

$$\Phi(u\mathbf{1}) = a(1 + p/u) + \kappa^2 p^2 (u-1)^2, \quad \frac{d}{du} \Phi(u\mathbf{1}) = -\frac{ap}{u^2} + 2\kappa^2 p^2 (u-1),$$

which is strictly increasing, negative on $(0, 1]$, and unbounded above, so the unique root solves $2p\kappa^2 u^2(u-1) = a$ and lies in $(1, \infty)$. This is the scalar cubic with κ^2 replaced by $p\kappa^2$, so the comparative statics in n and κ follow from Theorem 2, and the same implicit-function computation with $\partial F / \partial p > 0$ (through the $p\kappa^2$ factor) gives strict decrease in p . Block V6a confirms by free minimization over eigenvalues from thirty random anisotropic starts per configuration (gap to the isotropic value at most 2×10^{-14} ; maximal anisotropy of the argmin 9×10^{-7}), and block V6b confirms the p -path 1.4329, 1.2739, 1.2043, 1.1374, 1.0767 at $n = 50$, $\kappa = 0.15$. $\square$

D.3. The complete first-order total risk (Proposition 3). Fix the Gaussian scale family $x \sim \mathbf{N}(0, u)$, curvature split $c_1 = \kappa/2 = -c_0$, unit-variance Gaussian target, residual scale σ , and per-arm size n . Three objects enter the total risk of the OLS contrast (d_0, d_1) against the truth $\tau(x) = \kappa x^2$:

(a) *Sandwich variance around the realized projection.* With regression residual $r_a = \varepsilon_a + c_a(x^2 - u)$, the bread is $\text{diag}(1, u)$ and the meat is $\mathbb{E}(r_a^2 \mathbf{z} \mathbf{z}^\top) = \text{diag}(\sigma^2 + 2c_a^2 u^2, \sigma^2 u + 10c_a^2 u^3)$ for $\mathbf{z} = (1, x)^\top$, using $\mathbb{E}\{x^2(x^2 - u)^2\} = \mathbb{E}x^6 - 2u\mathbb{E}x^4 + u^2\mathbb{E}x^2 = 15u^3 - 6u^3 + u^3 = 10u^3$. Sandwiching and integrating over the unit-variance target

gives per-arm contributions $(\sigma^2 + 2c_a^2 u^2)/n$ from the intercept and $(\sigma^2 u + 10c_a^2 u^3)/(u^2 n)$ from the slope; summing the arms yields

$$\frac{2\sigma^2(1 + 1/u)}{n} + \frac{\kappa^2(u^2 + 5u)}{n}.$$

(b) *Projection drift.* The realized projection intercept is biased relative to the population projection. Writing sample moments m_k , the arm- a OLS intercept against the curved truth has expectation

$$\mathbb{E}(b_{a0}) = c_a \mathbb{E} \left[\frac{m_2^2 - m_1 m_3}{m_2 - m_1^2} \right] = c_a \left(u - \frac{2u}{n} \right) + O(n^{-2}),$$

by expanding in the $O_p(n^{-1/2})$ moments m_1, m_3 and $m_2 - u$, with $\mathbb{E}(m_1^2) = u/n$ and $\mathbb{E}(m_1 m_3) = 3u^2/n$; the slope drift is $O(n^{-2})$ by symmetry. The contrast intercept drift is therefore $\mathbb{E}(e_0) = -2\kappa u/n + O(n^{-2})$ (verified directly: simulation gives -0.0599 against the formula's -0.0600 at $n = 100$, $u = 4$, $c = 0.75$). Crossing the drift with the population bias $\kappa(u - 1)$ in the target integral contributes

$$2 \mathbb{E}(e_0) \kappa(u - 1) = -\frac{4\kappa^2 u(u - 1)}{n} + O(n^{-2}).$$

(c) *Population bias.* $\kappa^2\{(u - 1)^2 + \text{Var}_T(x^2)\}$ with $\text{Var}_T(x^2) = 2$ for the Gaussian target.

Adding (a) through (c) and simplifying the curvature part, $\kappa^2(u^2 + 5u) - 4\kappa^2 u(u - 1) = 3\kappa^2 u(3 - u)$, gives Proposition 3:

$$R(u) = \kappa^2\{(u - 1)^2 + 2\} + \frac{2\sigma^2(1 + 1/u) + 3\kappa^2 u(3 - u)}{n} + O(n^{-2}). \quad (\text{D.3})$$

For $n > 3$, the displayed first-order expression is coercive and strictly convex because

$$R''(u) = 2\kappa^2 \left(1 - \frac{3}{n} \right) + \frac{4\sigma^2}{nu^3} > 0, \quad R'(1) = \frac{3\kappa^2 - 2\sigma^2}{n}.$$

Its unique optimizer is therefore above one if $\kappa^2 < 2\sigma^2/3$, equal to one at equality, and below one if $\kappa^2 > 2\sigma^2/3$. Block V5f verifies this crossing: at $n = 50$, $\sigma = 1$, the optima are 0.9946 and 0.9898 for $\kappa = 0.9$ and 1, while the criterion optima remain 1.0236 and 1.0193. Block V5d validates (D.3) against 30,000-replication Monte Carlo at $n \in \{50, 200, 800\}$ and $u \in \{1, 2, 4\}$: maximal relative deviations 0.0013, 0.0001, 0.0005, at the Monte Carlo noise floor from $n = 200$; the sandwich-only formula without term (b) deviates by 0.087, 0.022, 0.005, decreasing approximately as $1/n$, which identifies the drift as a genuine first-order term. Block V5e records the default $\kappa = 0.15$ path

1.758, 1.430, 1.160, 1.040 at $n = 20, 50, 200, 1000$; it is a local sensitivity result, not a universal overdispersion conclusion.

Proposition D.1 (scale family). *Let the source be $x = \sqrt{u}Z$ with fixed symmetric Z , $\mathbb{E}Z^2 = 1$, and $\mathbb{E}Z^6 < \infty$. The sandwich addition beyond the noise part is, per arm, $c_a^2\{u^2q_Z + uk_Z\}/n$, where $q_Z = \mathbb{E}(Z^2 - 1)^2$ and $k_Z = \mathbb{E}Z^2(Z^2 - 1)^2$. It is nondecreasing in $u > 0$ for every such law, and it is strictly increasing if and only if $c_a \neq 0$ and $\Pr(Z^2 \neq 1) > 0$.*

Proof. With $x^2 - u = u(Z^2 - 1)$, the intercept meat term is $c_a^2\mathbb{E}(x^2 - u)^2 = c_a^2u^2q_Z$ and the slope meat term is $c_a^2\mathbb{E}\{x^2(x^2 - u)^2\}/u^2 \cdot u = c_a^2uk_Z$ after sandwiching, exactly as in (a). Both q_Z and k_Z are nonnegative. If $\Pr(Z^2 \neq 1) > 0$, then $q_Z > 0$, so the derivative $c_a^2(2uq_Z + k_Z)/n$ is positive for $c_a \neq 0$ and $u > 0$. Conversely, if $c_a = 0$ the addition vanishes, and if $\Pr(Z^2 \neq 1) = 0$, then $q_Z = k_Z = 0$ and it again vanishes. Gaussian Z gives $(q_Z, k_Z) = (2, 10)$; uniform gives $(0.8, 1.257)$; a scaled t_7 gives $(4, 116)$; and Rademacher $Z = \pm 1$ gives $(0, 0)$ exactly (blocks V7a–d). $\square$

D.4. The noise rule (Corollary 1). With profile $\sigma^2(u) = 1 + g(u - 1)$ and objective $h(u) = \sigma^2(u)(1 + 1/u)$ on the domain where the profile is positive,

$$h'(u) = g \left(1 + \frac{1}{u} \right) - \frac{1 + g(u - 1)}{u^2} = g - \frac{1 - g}{u^2},$$

which for $0 < g < 1$ has the unique root $u = \sqrt{(1 - g)/g}$, a minimum since h' is increasing. At $g = 0$, $h' < 0$ everywhere, so the objective decreases without bound; as $g \uparrow 1$ the root tends to zero. The rule sits above the representative spread for $g < 1/2$ and below it for $g > 1/2$ (block V8).

Appendix E. Computational Verification and Simulation Settings

E.1. Evidence taxonomy. Every block carries one of the shared program labels: **proof** (claim proved here; block is a consistency companion), **quadrature** (deterministic assertion, no Monte Carlo error), **independent** (independent numerical derivation against a separately derived closed form), **mc** (Monte Carlo with stated tolerance), **regression** (same-implementation check), **descriptive** (reported number, no assertion).

E.2. Verification map. Table E.1 lists every verification block with its evidence label and the statement it checks.

TABLE E.1. Verification blocks (suite 01_verification.R, seed 20260820; zero failures on the shipped run).

Block	Evidence	Statement checked
V1	independent	Lemma 1 decomposition over 50 random configurations (max diff 2.6×10^{-13})
V2	independent	Mean matching recovered by free minimization over the source mean (max dev 7×10^{-7})
V3	independent	Square-root optimum against free Cholesky optimization; efficiency ratio endpoints; budget monotonicity
V4	independent / proof / regression	Pairwise Frank–Wolfe + polish (gap 1.9×10^{-8}); optimum support 4; forced dense input reduces $30 \rightarrow 6$ with moment error 4.4×10^{-16} , simplex error zero, and nonnegative weights; optimal 1.51 vs equal 2.27, best-single 2.63, nearest-5 3.67
V5a–c	independent	Scalar criterion: single sign change; cubic residual 10^{-13} ; u^* paths $1.74 \rightarrow 1.00$ in n and decreasing in κ
V5d	mc	Complete first-order risk vs 30,000-rep MC at $n \in \{50, 200, 800\}$, $u \in \{1, 2, 4\}$ (max rel dev 0.0013/0.0001/0.0005); sandwich-only deviations 0.087/0.022/0.005 decrease approximately as $1/n$, as expected when first-order drift is omitted
V5e	independent	Default $\kappa = .15$ complete-risk optimizer path above one and toward one (1.758/1.430/1.160/1.040)
V5f	proof / independent	Complete-risk crossing at $\kappa^2 = 2\sigma^2/3$; at $n = 50$, $u^* = .9946/.9898$ for $\kappa = .9/1$
V6	independent / proof	Criterion- F optimum isotropic (30 restarts; gap $\leq 2 \times 10^{-14}$); u^* decreasing in p ; saturated aligned contamination attains both Gaussian-target bias terms, including zero-mismatch eigenspaces
V7	independent / quadrature / proof	Scale-family moments: Gaussian (2, 10) by MC; t_7 (4, 116) by quadrature; additions nondecreasing, strict iff $c_a \neq 0$ and $\Pr(Z^2 \neq 1) > 0$; Rademacher (0, 0) equality case
V8	independent / proof	Noise rule equals $\sqrt{(1-g)/g}$ on $0 < g < 1$; boundary behavior
V9	mc / independent / regression	Tail ordering with explicit protocols; redraw counts 13/0/0/0/0/0 at $n = 12/20/28/40/60/100$; balanced factorial exact when n is divisible by 2^p ; exact spread-shift formula $a_n + b_n p/u$ against MC

E.3. **Claim boundaries.** Table E.2 condenses, claim by claim, what the suite supports, under which conditions, and what is deliberately not claimed.

E.4. **Simulation settings.** Site pool: $K = 30$ sites, $p = 2$; site means drawn $N_2(\mathbf{0}, 1.1^2 \mathbf{I})$; site covariances are random rotations of $\text{diag}(U(0.15, 1.4))$; target mean $(0.3, -0.2)$ with identity covariance; seed 20260821. Frank–Wolfe uses 30,000 pairwise iterations with line search, followed by active-set polish and full-nullspace

TABLE E.2. What the verification suite supports and what it does not.
Block identifiers refer to the verification map of Table E.1.

Claim	Status
Leverage decomposition into intercept, Mahalanobis, and dispersion terms	Supported (V1)
Mean matching optimal at fixed source covariance	Supported (V2); fixed-covariance scope stated
Square-root allocation, efficiency ratio, and budget monotonicity	Supported (V3)
Convex site program; equivalence gap 1.9×10^{-8} ; polished optimum has support 4; dense 30-site input reduces to 6 with moment error 4.4×10^{-16} , unit mass, and nonnegative weights	Supported (V4; bound $q(q+1)/2 = 6$)
Scalar interior optimum and comparative statics	Supported (V5a–c)
Complete first-order total risk, including the projection-drift term; sandwich-only formula fails the convergence test	Supported (V5d)
Complete-risk optimizer: default path above 1; crossing below 1 iff $\kappa^2 > 2\sigma^2/3$	Supported (V5e–f; $\kappa = .9, 1$ included)
Criterion F : unique optimum $u^* \mathbf{I}$, decreasing in n, κ, p	Supported (V6; zero-third-moment source; Gaussian target for joint population-bias worst case)
Scale-family variance additions nondecreasing in u , strictly increasing iff $c \neq 0$ and $\Pr(Z^2 \neq 1) > 0$; Rademacher equality case; t7 moments by quadrature	Supported (V7)
Noise profile rule on its stated domain $0 < g < 1$	Supported (V8)
Tail ordering with explicit protocols; balanced factorial used only at valid multiples of 2^p ; redraw counts 13, 0, 0, 0, 0, 0 reported	Supported (V9a, V9b, V9d)
Exact spread-shift formula $a_n + b_n p/u$	Supported (V9c)
Comparator descriptions and publication status	Source-verified; no head-to-head performance claim
Leverage-optimal equals risk-optimal under differing noise	Not claimed; Corollary 1
Complete total-risk minimax from a contrast-curvature bound; robustness beyond quadratic curvature	Not claimed; arm curvatures need separate control; complementary methods cited
Site participation behavior, cost heterogeneity beyond noise, within-site clustering	Outside scope
Any relaxation of the residual floor or its non-identified correlation	Impossible in this framework; inherited from the companion

Carathéodory reduction. Scalar and multivariate criteria use closed forms; free minimizations use BFGS in log-eigenvalue coordinates with thirty starts. Complete-risk Monte Carlo uses 30,000 replications per cell, seed 20260822. Finite-sample tails use 8,000 replications per cell over $n \in \{12, 20, 28, 40, 60, 100\}$, with Gaussian, uniform, i.i.d. two-point, and balanced factorial two-point sources, seed 20260826. Two protocol details matter for the comparison in the main text. First, an i.i.d. two-point design can be singular at small n , so such draws are rejected and redrawn; the redraw counts across the six sizes were 13, 0, 0, 0, 0, 0, so the $n = 12$ cell is conditional on the redraw protocol while every larger size is not (the count at $n = 20$ was zero in all 8,000 replications). Second, the balanced factorial protocol is defined only when $n \equiv 0 \pmod{2^p}$; every displayed $p = 2$ size satisfies this condition, so the balanced curve is unconditional at those sizes. The exact spread-shift check uses seed 20260827 and 20,000 replications per cell. Software: R 4.6.0; no Markov chain Monte Carlo. The suite writes `verification_report.md` and `key_numbers.csv` and exits nonzero on failure.

E.5. Supplementary figure. Figure E.1 displays the scale-family variance additions of Proposition D.1, the closed forms behind blocks V7a through V7d.

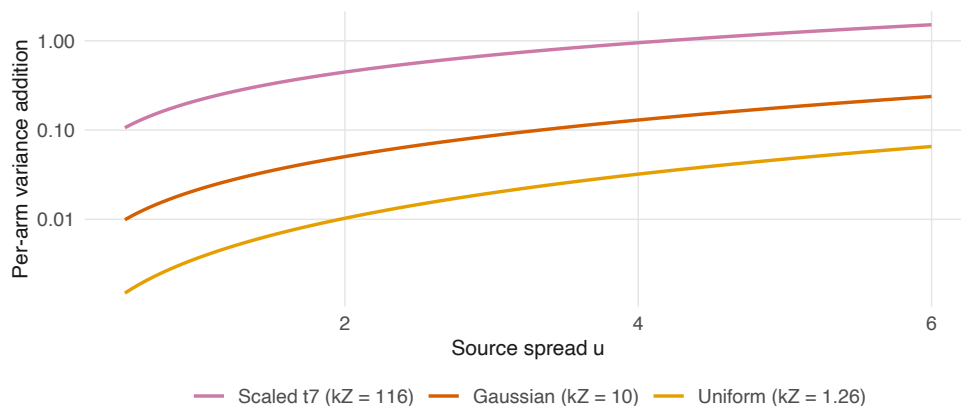


FIGURE E.1. Scale-family variance additions (Proposition D.1) for the three positive-addition Z laws at $c_a = 0.3$, $n = 50$, log scale. Each shown curve increases in u ; heavier tails raise the level through k_Z (10 for Gaussian, 116 for scaled t_7). The Rademacher edge case has $q_Z = k_Z = 0$ and is identically zero, so it cannot appear on the log scale.